EL NIÑO ACTION BRIEFING · 2 OCTOBER 2026 · VERIFIED RUN

East Portland, Oregon, USA

The people who die here are older and alone at home, whether the cause is 116 °F heat or an ice storm. Use this winter to build a named list of them, give everyone on it a working heat pump and a person who will knock, and keep the winter plan running despite El Niño's milder odds.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not resolved. Multnomah County (~795,000); East Portland is ~28% of the city, east of 82nd Ave (2010-era data)	United States ~3,500 for the study period. No Portland figure; the remaining Sep–Feb months are winter here	Study period Jun 2026 – Feb 2027	Winter cold snaps, ice storms and outages (now to Feb); then summer 2027 heat and wildfire smoke

13 recommendations: 1 confirmed, 9 corrected, 3 unverified, 0 contraindicated; five widely circulated practices are listed under "Do not do this". **Scope:** asked for "Portland, Oregon", this report covers East Portland, the neighbourhoods east of I-205 (Hazelwood, Lents, Powellhurst-Gilbert, Mill Park, Centennial, Glenfair, Wilkes, Parkrose). That is where the 2021 heat-dome response focused and where Portland's lower-income, most diverse residents now live. Rockwood (in Gresham, next door) is included where the evidence names it. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. **For Portland it says almost**

EXECUTIVE SUMMARY — AT A GLANCE



45.52°N 122.56°W, United States of America · Natural Earth

- **The risk.** The people who die here are older and alone at home, in heat or ice: the June 2021 heat dome killed 72 in Multnomah County, 71% of them living alone and 94% at home.
- **First action.** The City of Portland and Multnomah County get a working heat pump into every East Portland home of an older person living alone, including Rockwood, before June 2027.
- **The money.** $150,000 floor, $475,000 target, $525,000 ceiling; philanthropy's gap is people who check on people, Rockwood, and keeping units working.
- **Do not.** Do not hand out fans as the heat response: for older adults they become less effective above 33 °C and can worsen heat strain above 40 °C.*

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* Suggested formulation: check with a health worker before sharing.

nothing useful: any heat deaths in the study period fell in summer 2026, and September–February is winter here. This report is driven by the local record instead.

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected lives saved per unit of effort through this winter and before summer 2027. Backing lenses are shown per action. The RACI chart below assigns each one.

1. Get a working heat pump into every East Portland home of an older person living alone before June 2027, including Rockwood, which the city programme can't reach

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY · HISTORIAN · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
City of Portland Bureau of Planning and Sustainability (Portland Clean Energy Fund) via Earth Advantage and its Cooling Portland partners; Multnomah County for Rockwood	Waitlist cleared by 31 May 2027	~$1,030 per household at bulk prices (Dec 2024 expansion: $10.3m for 10,000 households)	Deaths at home in heat waves, and in winter outages, since a heat pump also heats

In the June 2021 heat dome, Portland reached 116 °F and 72 people died in Multnomah County: 78% were aged 60 or over, 71% lived alone, and 94% died at home. Of the death investigations that mentioned a cooling unit, 7 of 10 found it broken or unplugged (Multnomah County). Working home air conditioning cut the odds of heat death to 0.23 (95% CI 0.1–0.6) in a meta-analysis (Bouchama et al. 2007). The city's Cooling Portland programme has delivered more than 20,500 portable heat pumps and closed to applications on 30 June 2026 with a waitlist that prioritises people over 60, medically at risk, and living alone.

* Suggested formulation: check with a health worker before sharing.

Corrections: Oregon's Medicaid climate-device benefit (air conditioners, heaters, filters, through Health Share of Oregon) is real but limited to members in a qualifying transition, such as leaving homelessness or incarceration, who also have a climate-sensitive condition. It is a route for some, not the main route for isolated older renters. Cooling Portland covers the City of Portland only; the county's last air-conditioner programme for others (FY2023) installed about 180 units. The ~$38m programme total that one lens cited could not be verified.

2. Build a named list now of older people living alone, and on every heat or ice alert escalate from a phone call to a knock on the door and an escort to a cool or warm place

PARTIALLY CONFIRMED | FRONTLINE · SKEPTIC · EMERGENCY · PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Multnomah County Health Department; APANO and Latino Network (Cooling Portland partners) for their communities; IRCO, Unite Oregon and Rosewood consulted	**List built in October; live before the first ice event**	**$150,000 placement (see funding)**	**People dying alone and unseen, the 2021 pattern**

In 2021 the county phoned 3,850 older adults and handed out fans, and more than 60 people still died. The county's own action plan then called for door-to-door wellness checks. Social contact cut the odds of heat death to 0.40, and not leaving home daily raised them to 3.35 (Bouchama 2007). Start the list from Cooling Portland recipients and the community organisations' senior clients, and have volunteers who speak each person's language call, then knock if there's no answer. The same list serves this winter's ice storms and next summer's heat.

* Suggested formulation: check with a health worker before sharing.

Corrections: a call on its own is not enough; the action requires escalation to a visit. No existing county check-in registry was found to build on, and Aging, Disability and Veterans Services lists only its helpline. IRCO, Unite Oregon and Rosewood are not Cooling Portland partners, so they are consulted, not responsible. Rosewood's site still shows a COVID-era closure notice; confirm it is operating.

3. Check in May 2027 that the installed units actually work

PARTIALLY CONFIRMED | SKEPTIC · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Cooling Portland partners (APANO, Latino Network, EMO, NW Native Chamber, SACOO, African American Alliance for Homeownership; housing providers Home Forward, NW Housing Alternatives, REACH CDC)	May 2027	$100,000 placement (see funding)	A unit that is broken, unplugged or feared for its bill on the hottest day

In 2021, 7 of 10 cooling units noted in death investigations were broken or unplugged. Counting units delivered is not protection. Visit or call each recipient in May, test the unit, and pair it with the bill help in action 4.

Correction: the programme already pays partners to install, so fund post-programme testing, not waitlist set-up. No public data exist on whether delivered units are used.

4. In October, enrol households in PGE's income-qualified bill discount and federal energy assistance

CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
PGE; Multnomah County Department of County Human Services (LIHEAP and Oregon Energy Assistance)	October–November 2026	Staff time; benefits paid by PGE ratepayers and federal HHS	People not running heating or cooling for fear of the bill

* Suggested formulation: check with a health worker before sharing.

PGE's discount is 15–80% off energy charges, needs no financial documents, and lasts two years. LIHEAP and Oregon Energy Assistance cover households at or below 60% of state median income; the programme year opened 1 October 2026. Oregon already bars residential disconnection when the forecast high is below 32 °F or during an NWS heat advisory (OAR 860-021-0407), so there is no need to campaign for that.

5. Keep the winter plan running: shelters on the county's thresholds, a bus fallback when MAX stops, and carbon monoxide safety during outages

PARTIALLY CONFIRMED | EMERGENCY · SKEPTIC · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Multnomah County Emergency Management and Homeless Services Department; TriMet; Portland Bureau of Emergency Management	1 Dec 2026	Existing severe-weather budget; $75,000 placement for East Portland outreach (unverified)	Hypothermia, carbon monoxide poisoning, and people stranded when transit stops

NOAA favours a warmer Northwest winter but gives no strong dry signal for Portland, and warmer odds shift the average, not the extremes. The January 2024 storms came during an El Niño (ONI +2.0): MAX stopped for three days, the county logged four suspected hypothermia deaths, and three people died when a power line fell on a car. In the February 2021 ice storm, more than 270,000 people lost power and four Oregonians died of carbon monoxide poisoning while trying to stay warm. Severe-weather shelters open when 25 °F or below, 1 inch of snow, or 32 °F with 1 inch of driving rain is forecast for four hours or more overnight. Plan warming sites east of I-205 with backup power, buses for when trains stop, and carbon monoxide alarms and generator warnings in several languages. Carry naloxone at warming sites: most deaths among unhoused people are overdoses (282 of 394 medical-examiner cases in 2023).*

* Suggested formulation: check with a health worker before sharing.

Corrections: the four 2024 hypothermia deaths were in inner N, NE and SE Portland ZIP codes, not East Portland, so the East Portland cold-death link is inferred. The car deaths belong to the 16–19 January ice storm, not the 13–16 January storm. The naloxone point is an inference without an effect size.

6. Smoke readiness for households with lung or heart disease who don't qualify for Medicaid devices

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Multnomah County Health Department with Cooling Portland partners	Before July 2027	$50,000 placement (see funding)	Respiratory and cardiac harm during wildfire smoke

Low snowpack after an El Niño winter raises summer drought and fire risk. In September 2020 Oregon's air quality index passed 500. Portable air cleaners cut indoor fine particles by about half in one study (PMC13481836) and by 15% in a field study during a Los Angeles fire (PMC12804073). Medicaid covers filters only for the narrow transition groups, so the gap is real.*

No mortality effect size exists; this reduces illness more than deaths. Supply replacement filters, not just units.

Unverified: trim trees around East Portland power lines and push downed-line safety messages

UNVERIFIED

PGE vegetation management and "stay in your car, call 911" messaging through PublicAlerts before winter. Falling trees and lines killed people in 2024, but this action, PGE's budget, and East Portland's share of tree-caused outages were not verified.*

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	City BPS / Clean Energy Fund	Earth Advantage & Cooling Portland partners	County Health Dept	County Emergency Mgmt	County Homeless Services	County Human Services (energy aid)	Portland Bureau of Emergency Mgmt	Health Share of Oregon	PGE	TriMet	IRCO, Unite Oregon, Rosewood	NWS Portland	City of Gresham (Rockwood)
1. Heat pumps for older people living alone	**A**	**R**	**C**					**C**	**I**		**I**		**C**
2. Named list, knock and escort on alerts	**I**	**R**	**A**	**C**			**R**			**C**	**C**	**C**	**I**
3. May check that units work	**A**	**R**	**C**						**I**				
4. Bill discount and energy assistance enrolment		**R**				**A**			**R**		**I**		
5. Winter shelters, buses, carbon monoxide safety		**I**	**C**	**A**	**R**		**R**		**C**	**R**	**I**	**C**	**I**
6. Smoke readiness		**R**	**A**					**C**			**I**	**C**	
Unverified: trees and downed lines				**I**			**C**		**A/R**				

What the chart shows. **The Cooling Portland partners do the work on five of seven rows**, so whether that network keeps operating after the programme ends in 2026 is the plan's biggest dependency. **The County Health Department is accountable for the two actions that rest on people, not hardware. Rockwood has no owner with money behind it**: the City of Gresham is only consulted because no Gresham cooling programme was found.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. Public money already covers the hardware (Cooling Portland) and the bills (PGE discount, LIHEAP); philanthropy's gap is people who check on people, Rockwood, and keeping units working.

FLOOR

$150,000 (wellness checks)

TARGET

$475,000

CEILING

$525,000

LARGEST UNCERTAINTY

Whether a 2021-scale heat event comes in summer 2027; in a typical year the county records no heat deaths

* Suggested formulation: check with a health worker before sharing.

Target = $150,000 + $100,000 + $150,000 + $75,000. Ceiling = target + $50,000 smoke readiness. The funder lens proposed a $575,000 target and a $900,000 ceiling; verification cut the partner-grant line from $200,000 to $100,000 because Cooling Portland already pays partners to install (–17%, under the one-third threshold), and removed $275,000 of ceiling that had no placement behind it.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
2–3 culturally specific community organisations (APANO and Latino Network confirmed partners; IRCO senior services)	Action 2: named list, calls, door knocks, escorts	$150,000	Grants	Social contact OR 0.40 (Bouchama 2007); Philadelphia warning system ~117 deaths averted for ~$210,000 (Ebi et al. 2004, as reported in PMC7618602; original not reachable)	Partially confirmed; Rosewood as anchor unconfirmed
Cooling Portland partners	Action 3: test installed units after the programme ends	$100,000	Staff grants	7 of 10 units in 2021 death investigations broken or unplugged	Corrected (was $200,000 incl. waitlist set-up, which duplicates programme pay)
A Rockwood organisation or Multnomah County (Rockwood, Gresham)	Action 1 outside the city boundary	$150,000	Direct grant for units	~$1,030 per household at Cooling Portland's bulk price; a small purchase will cost more, so ~100–145 households	Partially confirmed; no Gresham programme found
County outreach teams or East Portland organisations	Action 5: winter outreach, transport, carbon monoxide safety	$75,000	Grant	No published cost per death	Unverified; county severe-weather budget not reached
Cooling Portland partners with County Health	Action 6: air purifiers and replacement filters	$50,000 (ceiling only)	In-kind	Indoor PM2.5 cut 15–51%; no mortality effect	Partially confirmed; gap real

* Suggested formulation: check with a health worker before sharing.

The funder's one thing: fund people who check on people, not more hardware. The public sector has already bought more than 20,500 units. People died alone at home, often beside a unit that wasn't running.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would cost lives here.

Handing out fans as the heat response*

CONTRAINDICATED

For older adults, fans become less effective above 33 °C and can worsen heat strain above 40 °C (O'Connor, Kenny and Meade, *CMAJ* 2025); agencies should advise against fans above 35 °C (Meade et al. 2024). Portland reached 46.7 °C (116 °F) in 2021, and the county's response then was hundreds of fans.

Instead: working heat pumps or air conditioning, a cool room, wetting skin and clothing, and escorts to cooled places.

Counting cooling and warming centres as protection

CONTRAINDICATED

In August 2023, three centres had nearly 1,000 visits over four days in a county of about 800,000. In 2021, 211info dropped more than 750 calls about shelters and transport, and TriMet suspended trains. The county itself says cooling shelters "don't work for everyone who is at risk".

Instead: bring protection to the home (actions 1 and 2), and treat centres as a supplement with transport.

Relying on phone calls alone

CONTRAINDICATED

In 2021 the county called 3,850 older adults and more than 60 people still died.

* Suggested formulation: check with a health worker before sharing.

Instead: calls that escalate to a knock and an escort when nobody answers or the home is hot.

Standing down the winter plan because El Niño means a mild winter

CONTRAINDICATED

The January 2024 storms, with their deaths and three-day transit shutdown, came during an El Niño of similar strength. Seasonal odds don't remove arctic outbreaks through the Columbia Gorge, and East Portland sits closest to the Gorge.

Instead: trigger shelters and transport on the county's temperature thresholds, whatever the seasonal outlook says.

Counting units delivered as success, or running heaters and generators indoors *

CONTRAINDICATED

Units that are broken, unplugged or unaffordable to run protect nobody (2021 death investigations). Indoor generators and grills killed four Oregonians during the February 2021 outage.

Instead: a May check that every unit runs, bill help (action 4), and carbon monoxide alarms with multilingual generator warnings.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Sep 2020	Labor Day wildfires; smoke over Portland	Oregon air quality index above 500	Not documented
Feb 2021	Ice storm	Over 270,000 people without power, the largest outage in state history; 4 Oregon carbon monoxide deaths	Not documented

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Jun 2021	Heat dome, 116 °F	72 deaths in Multnomah County: 78% aged 60+, 71% living alone, 94% at home	Oregon SB 1536 (landlords can't ban portable cooling; new units need cooling); Cooling Portland (2022)
Aug 2023	Heat wave	Nearly 1,000 cooling-centre visits in four days	Showed low centre uptake
Jan 2024 (El Niño, ONI +2.0)	Winter storms and ice	4 suspected hypothermia deaths in inner Portland; 3 killed by a fallen power line; MAX stopped 3 days	FEMA DR-4768: $40.5m to public agencies, no individual assistance
2023	Deaths among people experiencing homelessness	394 medical-examiner cases (456 including hospital records); 282 overdoses; 1 hypothermia	Not documented
2026	Cooling Portland closes to applications (30 Jun)	More than 20,500 units delivered by May	Waitlist only; programme ends in 2026

The recurrence nobody has noticed: heat and ice kill the same people here. The 2021 heat victims and the 2021 outage victims were alone at home, without a working way to cool or heat. A heat pump with a named person behind it answers both the hazard this El Niño winter is likely to bring and the one it may not.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Medicaid's climate-device benefit is the funded route to home cooling** (Public health) versus **it is limited to people in a qualifying transition** (Economist, confirmed in verification). Resolved: it serves a narrow group; Cooling Portland and Rockwood funding carry the rest.
- **Buy hardware** (Emergency manager, Historian) versus **fund people who check on people** (Philanthropist, Skeptic). Both are kept: public money covers hardware, philanthropy covers people and Rockwood.
- **El Niño means a mild winter** versus **the deadliest recent storms came in an El Niño** (Historian, Skeptic). The plan treats cold as live; the East Portland cold-death link itself is inferred.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED

BLIND SPOT

Two things. First, **whether delivered units are actually used**: no data exist on running costs, breakdowns or use among Cooling Portland's 20,500 recipients, and that decides whether the hardware already bought protects anyone. Second, **who owns this after 2026**: the city programme and its partner network wind down, and no one has been named to keep the list of people living alone, test the units, or cover Rockwood. Neither question is answered by El Niño research; both are answered by one meeting between the Bureau of Planning and Sustainability and the County Health Department.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** Angus asked for "Portland, Oregon". The report was scoped to East Portland (east of I-205) because that is where the 2021 heat-dome response focused and where the city's lower-income residents live; the choice was stated before the run, with an offer to change it. Climate Impact Lab impact region, as the report records it: not resolved. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.

* Suggested formulation: check with a health worker before sharing.

- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list. The RACI chart was built from the owners named in the verified actions.
- **Verification.** Five verifier agents checked the recommendations on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. Two verifiers disagreed on the 2021 death profile; the county's own news page confirmed it.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted: web search was exhausted for this session, and its download sits behind a sign-up form.
- **Health wording.** Passages where a wording error could hurt someone (cooling and fans, smoke and air cleaners, cold, carbon monoxide and generators, naloxone, and when to call 911) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. The ~$1,030 per household is the programme's own expansion figure ($10.3m for 10,000 households), labelled as a bulk price. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Run-specific gaps: web search unavailable; several multco.us and OPB pages returned 404 and oregonlive.com was blocked; neighbourhood-level death counts for East Portland, the county severe-weather budget, PBEM team locations, 211info services, Rockwood's 2021 over-representation, and use of delivered cooling units were not verified; the Philadelphia cost figure rests on a secondary source.

References · Heat: Multnomah County, 2021 heat deaths, 2021 preliminary findings, Aug 2023 cooling centres, Bouchama et al. 2007, O'Connor, Kenny & Meade 2025, Meade et al. 2024, Philadelphia HHWWS as reported, 2021 heat wave (confirmed) · Programmes: Cooling Portland, Cooling Portland 2026

* Suggested formulation: check with a health worker before sharing.

update, OHP climate supports, SB 1536, PGE bill discount, Oregon energy assistance, OAR 860-021-0407, County AC programme FY2023 (confirmed, corrected) · Winter: Jan 2024 storm, 2023–24 winter, Jan 2024 hypothermia deaths, Feb 2021 storm, severe-weather shelter thresholds, Domicile Unknown, FEMA DR-4768, CPC outlook, NOAA ONI (confirmed, corrected) · Smoke: air cleaners study, Los Angeles field study, 2020 fire season · Place: East Portland, Lents, Hazelwood, Springwater Corridor, Cooling Portland partners (confirmed)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 2 October 2026 for East Portland, Oregon. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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